# Analytic parameter sensitivities in ManifoldsX: scope and validation

manifoldsx.github.io/differentiable-dynamics

ManifoldsX computes how motion and joint loads change with spring and bushing stiffness and damping, analytically, at every step of a simulated trajectory. The derivatives cover consistent initialization, the implicit index-3 HHT step and the rotation update. They were compared with independent perturbed simulations over 30-second runs. These are numerical checks of the derivatives; they say nothing about hardware accuracy.

## 1. What they are for

### Understanding compliance

How a stiffness or damping coefficient changes displacement, velocity, acceleration and reaction forces at every instant.

### Calibration and design

Gradients are the building blocks for parameter fitting and design studies. A fitting or optimization workflow for a given application still has to be built and validated.

## 2. Validation

Table 1. Validation at solver revision b6e19fc

| Study | Evidence |
|---|---|
| Spring stiffness and damping | Analytic trajectory derivatives compared with Richardson-extrapolated finite differences over 30-second runs; every closed-form term also checked against finite differences. |
| 6D bushings | All 12 diagonal coefficients; 24 trajectories of 30 seconds on grounded and moving-parent connections. |
| Bushing motion derivatives | Worst sampled error relative to the peak reference: $1.11\times10^{-7}$ for position and rotation, $2.37\times10^{-7}$ for velocity. |
| Bushing accelerations and loads | Same metric: $3.80\times10^{-5}$ for acceleration, $5.47\times10^{-5}$ for multipliers and $5.46\times10^{-5}$ for generalized reactions. These are limited by the finite-difference reference, not by the derivatives. |
| Sparse sensitivity solve | Damped chains from 10 to 500 links, with one symbolic pattern analysis per run. |

Errors are normalized by the peak reference value over the sampled times, not per component at every instant. The analytic runs used no finite-difference evaluations; independent perturbed simulations supplied the reference. In the bushing tests, computing sensitivities left the nominal trajectory bit-for-bit unchanged.

## 3. Scope

### Covered

Forward sensitivities for spring stiffness and damping and the 12 diagonal bushing coefficients, one scalar parameter per sensitivity instance; bushing support is opt-in. The trajectory studies use index-3 HHT with full-rank revolute, spherical and translational joints and a parameter-independent initial pose and velocity.

### Not covered yet

Contact, controllers and runtime loads, redundant or rank-changing constraints, projection and other integrators. Adaptive step decisions are not differentiated. The Stewart motion benchmark does not establish differentiability of that whole model.

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